MATH 152 **QUIZ 2** Name Solutions

Mrs. Bonny Tighe 6.2-6.3

25 points TA_____ Wed 9/11/2019

6pts

1. Find the volume of the solid formed by rotating the given region about the specified line. Sketch the region and the solid and a representative **disc or washer**.

a) The region bounded by $y=(x+1)^2$ *and* $y=4$ about the x-axis.

$(x+1)^2=4$

$x+1=\pm 2$

$x=\pm 2-1$

$x=1,-3$

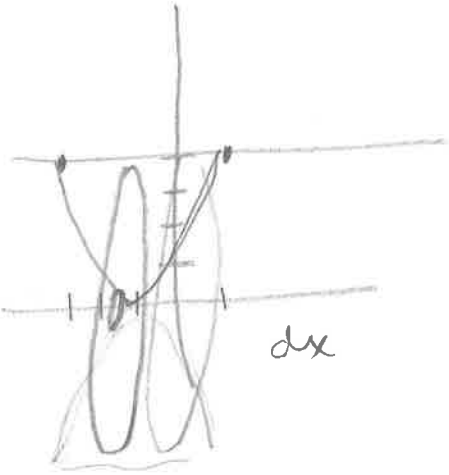
$$\pi\int_{-3}^{1} 4^2-((x+1)^2)^2\,dx$$

$$\pi\int_{-3}^{1} 16-(x+1)^4\,dx = \pi\left[16x-\tfrac{1}{5}(x+1)^5\right]_{-3}^{1}$$

$$\pi\left[16-\tfrac{1}{5}(2)^5-\left(16(-3)-\tfrac{1}{5}(-2)^5\right)\right]$$

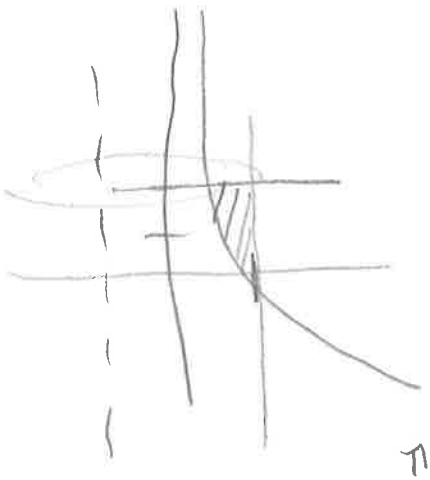
$$\pi\left[16-\tfrac{32}{5}+48-\tfrac{32}{5}\right]=\pi\left[64-\tfrac{64}{5}\right] \leftarrow \text{ok}$$

$$64\pi\left[1-\tfrac{1}{5}\right]=64\pi(4/5)=\frac{256\pi}{5}$$



6pts

b) The region bounded by $x=e^{-2y}$, $x=1$ *and* $y=2$ about the line $x=-1$



$$\pi\int_0^2 (1+1)^2-(e^{-2y}+1)^2\,dy$$

$$\pi\int_0^2 2^2-(e^{-4y}+2e^{-2y}+1)\,dy$$

$$\pi\int_0^2 (3-e^{-4y}-2e^{-2y})\,dy$$

$$\pi\left[3y+\tfrac{1}{4}e^{-4y}+e^{-2y}\right]_0^2$$

$$\pi\left[\left(6+\tfrac{1}{4}e^{-8}+e^{-4}\right)-\left(\tfrac{1}{4}e^0+e^0\right)\right]=$$

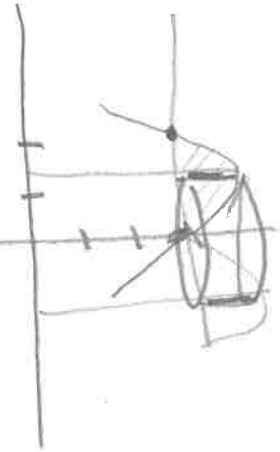
$$\pi\left[6+\tfrac{1}{4}e^{-8}+e^{-4}-\tfrac{1}{4}-1\right]=\pi\left[19/4+\tfrac{1}{4}e^{-8}+e^{-4}\right]$$

24

$3+2y-y^2=3 \qquad 2y-y^2=0 \qquad y=0,2$

6 pts 2. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line using the **cylindrical shells** method. Sketch the region and a representative cylinder.

a) $x=3+2y-y^2$ and $x=3$ rotated about the x-axis;



$$2\pi\int_0^2 y\left[3+2y-y^2-3\right]dy$$

$$2\pi\int_0^2 (2y^2-y^3)\,dy = 2\pi\left[\tfrac{2}{3}y^3-\tfrac{1}{4}y^4\right]_0^2$$

$$2\pi\left[\tfrac{2}{3}(8)-\tfrac{1}{4}(16)\right] = 2\pi\left[\tfrac{16}{3}-4\right]$$

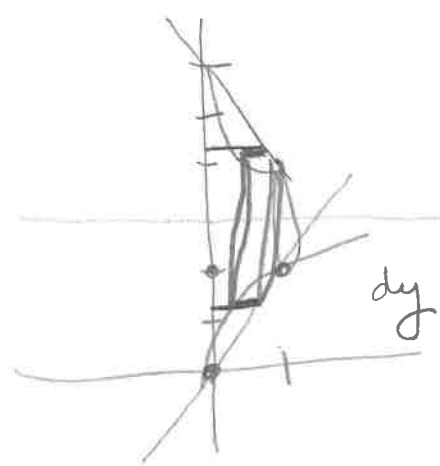
$$2\pi\left[\tfrac{16}{3}-\tfrac{12}{3}\right] = 2\pi\left[\tfrac{4}{3}\right] = 8\pi/3$$

7 pts b) The region bounded by $y=2\sqrt{x}$ *and* $y=2x$ about y = 3. using shells

$2\sqrt{x}=2x$

$x=0,1$

$\left(\tfrac{y}{2}\right)^2=x \qquad \tfrac{1}{2}y=x$



$$2\pi\int_0^2 (3-y)\left[\tfrac{1}{2}y-\tfrac{1}{4}y^2\right]dy =$$

$$2\pi\int_0^2 \left(\tfrac{3}{2}y-\tfrac{3}{4}y^2-\tfrac{1}{2}y^2+\tfrac{1}{4}y^3\right)dy$$

$$2\pi\int_0^2 \left(\tfrac{3}{2}y-\tfrac{5}{4}y^2+\tfrac{1}{4}y^3\right)dy =$$

$$2\pi\left[\tfrac{3}{4}y^2-\tfrac{5}{12}y^3+\tfrac{1}{16}y^4\right]_0^2 =$$

$$2\pi\left[\left(\tfrac{3}{4}(4)-\tfrac{5}{12}(8)+\tfrac{1}{16}(16)\right)-\right.$$

$$2\pi\left[3-\tfrac{10}{3}+1\right]$$

$$2\pi\left[4-\tfrac{10}{3}\right] = 2\pi\left(\tfrac{2}{3}\right) = \frac{4\pi}{3}$$

$-\tfrac{3}{4}+\tfrac{5}{12}-\tfrac{1}{16}$

$\dfrac{-36+20-3}{48}$ $-39+20$

MATH 152 **QUIZ 2A** Name Solutions

Mrs. Bonny Tighe 6.2-6.3

25 points TA_____Wed 9/11/2019

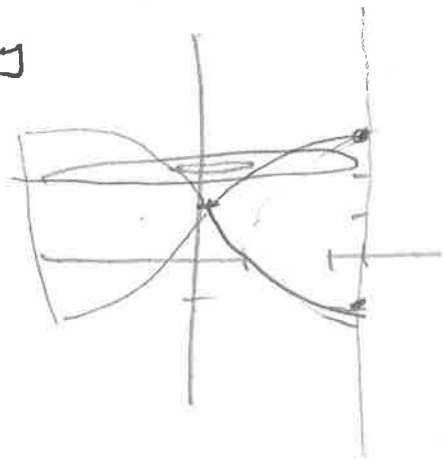
6pts

$(y-1)^2 = 4$
$y-1 = \pm 2$
$y = \pm 2 + 1$
$3, -1$

1. Find the volume of the solid formed by rotating the given region about the specified line. Sketch the region and the solid and a representative **disc or washer**.

a) The region bounded by $x = (y-1)^2$ *and* $x = 4$ about the y-axis. dy

$$\pi \int_{-1}^{3} 4^2 - ((y-1)^2)^2 \, dy = \pi \int_{-1}^{3} 16 - (y-1)^4 \, dy$$



$$\pi \left[16y - \tfrac{1}{5}(y-1)^5\right]_{-1}^{3} =$$

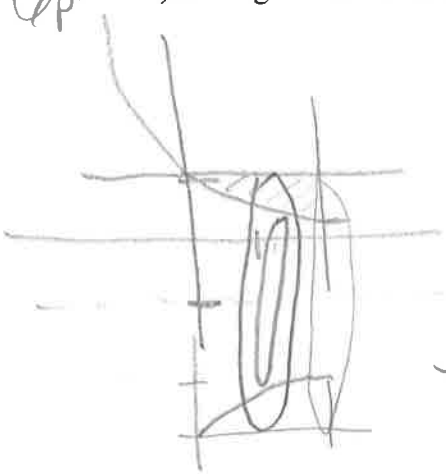
$$\pi \left[\left(48 - \tfrac{1}{5}(2)^5\right) - \left(-16 - \tfrac{1}{5}(-2)^5\right)\right]$$

$$\pi \left[48 - \tfrac{32}{5} + 16 - \tfrac{32}{5}\right] = \pi \left[64 - \tfrac{64}{5}\right]$$

$$64\pi \left[1 - \tfrac{1}{5}\right] = 64\pi(4/5) = \frac{256\pi}{5}$$

6pts

b) The region bounded by $y = e^{-2x}$, $x = 2$ *and* $y = 1$ about the line $y = -1$



$$\pi \int_0^2 (1+1)^2 - (e^{-2x}+1)^2 \, dx = \pi \int 4 - (e^{-2x}+1)^2 \, dx$$

$$\pi \int_0^2 4 - (e^{-4x} + 2e^{-2x} + 1) \, dx = \pi \int (3 - e^{-4x} - 2e^{-2x}) \, dx$$

$$\pi \left[3x + \tfrac{1}{4}e^{-4x} + e^{-2x}\right]_0^2$$

$$\pi \left[\left(6 + \tfrac{1}{4}e^{-8} + e^{-4}\right) - \left(0 + \tfrac{1}{4} + 1\right)\right] =$$

$$\pi \left[6 + \tfrac{1}{4}e^{-8} + e^{-4} - 5/4\right] = \pi \left[\tfrac{19}{4} + \tfrac{1}{4}e^{-8} + e^{-4}\right]$$

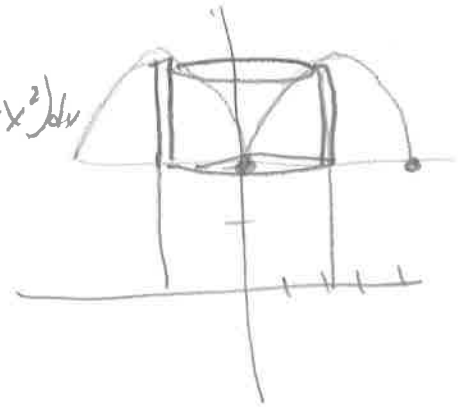
$2+4x-x^2=2 \qquad 4x-x^2=0$

$x=0,4$

6pts

2. Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line using the **cylindrical shells** method. Sketch the region and a representative cylinder. .

a) $y=2+4x-x^2,\ y=2$ rotated about the y-axis;



$$2\pi\int_0^4 x\left[2+4x-x^2-2\right]dx = 2\pi\int_0^4 x(4x-x^2)dx$$

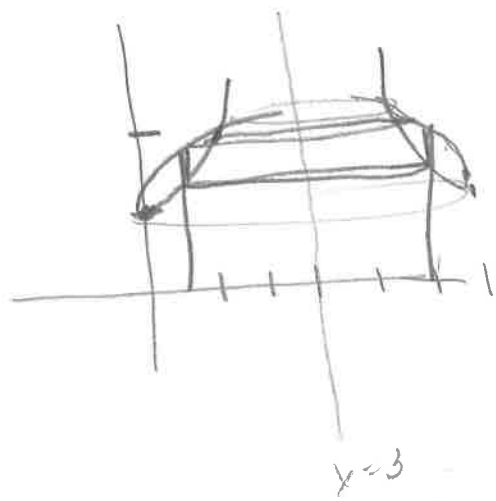
$$2\pi\int_0^4 (4x^2-x^3)dx = 2\pi\left[\tfrac{4}{3}x^3-\tfrac{1}{4}x^4\right]_0^4$$

$$2\pi\left[\tfrac{4}{3}(4)^3-\tfrac{1}{4}(4^4)\right] = 2\pi\, 4^4\left[\tfrac{1}{3}-\tfrac{1}{4}\right]$$

$$2\pi(4^4)\left(\tfrac{1}{12}\right) = 2\pi\frac{(64)}{3} = \frac{128\pi}{3}$$

7pts

b) The region bounded by $y=\sqrt{x}+1$ *and* $y=x^2+1$ about x = 3. using shells



$$2\pi\int_0^1 (3-x)\left[\sqrt{x}+1-(x^2+1)\right]dx$$

$$2\pi\int_0^1 (3-x)(\sqrt{x}-x^2)\,dx$$

$$2\pi\int_0^1 (3x^{1/2}-3x^2-x^{3/2}+x^3)dx$$

$$2\pi\left[3\cdot\tfrac{2}{3}x^{3/2}-x^3-\tfrac{2}{5}x^{5/2}+\tfrac{1}{4}x^4\right]_0^1$$

$$2\pi\left[2-1-\tfrac{2}{5}+\tfrac{1}{4}-0\right]$$

$$2\pi\left[1-\tfrac{2}{5}+\tfrac{1}{4}\right] = 2\pi\left[\frac{20-8+5}{20}\right] = 2\pi\left[\frac{17}{20}\right]$$

ok ↑

$$\frac{17\pi}{10}$$